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Web browser extension script obfuscation system

Abstract

A computing device comprising a web browser processes instructions to load, by the web browser, first executable code of a browser extension. The first executable code causes the web browser to load a file, wherein the file comprises file content and metadata. The web browser decodes second executable code from the metadata and processes the second executable code to extract obfuscated information embedded in the file content. The web browser extracts third executable code from the extracted obfuscated information and provides, to a user of the computing device, first browser extension functionality by processing the first executable code and second browser extension functionality by executing the third executable code.

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Background/Summary

BACKGROUND

(1) A web browser may be configured to render content from web pages and other sources. In some cases, web browser extensions and/or web browser applications may be installed in the web browser to provide additional functionality and features to enhance a user's browsing experience. For example, browser extensions may provide additional functionality and features based on the content from the web pages rendered in the web browser. When delivered, the browser extension code or scripts may be delivered in a format that may be easily accessible and/or understandable to a human user. As such, the browser extension code may be easily accessed and, thus, vulnerable to reverse engineering. It may be desirable to provide systems and techniques to control actions related to the browser extensions.

SUMMARY

(2) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosure. The summary is not an extensive overview of the disclosure. It is neither intended to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the description below.

(3) Aspects of the disclosure provide effective, efficient, scalable, and convenient technical

solutions that address and overcome the technical problems associated with obfuscating or otherwise providing executable code for scripting functions in a format not easily understood by a human.

(4) A computing device may process instructions of a web browser application, which in turn may processes instructions to load first executable code of a browser extension. The first executable code causes the web browser to load a file, wherein the file comprises file content and metadata. The web browser decodes second executable code from the metadata and processes the second executable code to extract obfuscated information embedded in the file content. The web browser extracts third executable code from the extracted obfuscated information and provides, to a user of the computing device, first browser extension functionality by processing the first executable code and second browser extension functionality by executing the third executable code.

(5) These features, along with many others, are discussed in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

(2) FIG. 1 shows an illustrative computing device processing obfuscated code for browser extensions in accordance with one or more aspects described herein;

(3) FIG. 2 shows an illustrative method for processing obfuscated code for browser extensions in accordance with one or more aspects described herein;

(4) FIG. 3 shows an illustrative operating environment in which various aspects of the disclosure may be implemented in accordance with one or more aspects described herein;

(5) FIG. 4 shows an illustrative block diagram of workstations and servers that may be used to implement the processes and functions of certain aspects of the present disclosure in accordance with one or more aspects described herein; and

(6) FIG. 5 shows an illustrative example of steganographic information included in an image.

DETAILED DESCRIPTION

(7) In the following description of various illustrative embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments in which aspects of the disclosure may be practiced. It is to be understood that other embodiments may be utilized, and structural and functional modifications may be made, without departing from the scope of the present disclosure.

(8) It is noted that various connections between elements are discussed in the following description. It is noted that these connections are general and, unless specified otherwise, may be direct or indirect, wired or wireless, and that the specification is not intended to be limiting in this respect.

(9) The above-described examples and arrangements are merely some example arrangements in which the systems described herein may be used. Various other arrangements employing aspects described herein may be used without departing from the invention.

(10) FIG. 1 shows an illustrative computing device processing obfuscated code for browser extensions in accordance with one or more aspects described herein. The computing system environment **100** may be used according to one or more illustrative embodiments. The computing system environment **100** is only one example of a suitable computing environment and is not intended to suggest any limitation as to the scope of use or functionality contained in the disclosure. The computing system environment **100** should not be interpreted as having any dependency or requirement relating to any one or combination of components shown in illustrative computing system environment **100**.

(11) The computing system environment **100** may include a computing device **101** having a

processor **103** for controlling overall operation of the computing device **101** and its associated components, including a Random Access Memory (RAM) **105**, a Read-Only Memory (ROM) **107**, a communications module **109**, and a memory **115**. The computing device **101** may include a variety of computer readable media. This computer readable media may be any available media that may be accessed by the computing device **101**, may be non-transitory, and may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, object code, data structures, program modules, or other data. Examples of computer readable media may include Random Access Memory (RAM), Read Only Memory (ROM), Electronically Erasable Programmable Read-Only Memory (EEPROM), flash memory or other memory technology, Compact Disk Read-Only Memory (CD-ROM), Digital Versatile Disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed by the computing device **101**.

(12) Although not required, various aspects described herein may be embodied as a method, a data transfer system, or as a computer-readable medium storing computer-executable instructions. For example, a computer-readable medium storing instructions to cause a processor to perform steps of a method in accordance with aspects of the disclosed embodiments is contemplated. For example, aspects of method steps disclosed herein may be executed on a processor on the computing device **101**. Such a processor may execute computer-executable instructions stored on a computer-readable medium.

(13) Software may be stored within the memory **115** and/or storage to provide instructions to the processor **103** for enabling the computing device **101** to perform various functions as discussed herein. For example, the memory **115** may store software used by the computing device **101**, such as an operating system **117**, one or more application programs **119**, API functionality, and at least one associated database **121**. In addition, some or all of the computer executable instructions for the computing device **101** may be embodied in hardware or firmware. Although not shown, the RAM **105** may include one or more applications representing the application data stored in the RAM **105** while the computing device **101** is on and corresponding software applications (e.g., software tasks) are running on the computing device **101**.

(14) The communications module **109** may include a microphone, keypad, touch screen, and/or stylus through which a user of the computing device **101** may provide input, and may also include one or more of a speaker for providing audio output and a video display device for providing textual, audiovisual and/or graphical output. The computing device **100** may also include one or more optical scanners (not shown).

(15) The computing device **101** may operate in a networked environment supporting connections to one or more remote computing devices and/or computing systems, such as an internal network environment (e.g., an intranet **133**) or an external computing network (e.g., the internet **131**). In some cases, computing devices (e.g., hosts, servers, and the like) may provide content accessible via an application program of the one or more applications **119** or via a web browser **140**.

(16) The network connections depicted in FIG. **1** may include a Local Area Network (LAN) **125** and/or a Wide Area Network (WAN) **129**, as well as other networks. When used in a LAN networking environment, the computing device **101** may be connected to the LAN **125** through a network interface or adapter in the communications module **109**. When used in a WAN networking environment, the computing device **101** may include a modem in the communications module **109** or other means for establishing communications over the WAN **129**, such as via a network **131** (e.g., public network, private network, Internet **131**, intranet **133**, and the like). The network connections shown are illustrative and other means of establishing a communications link between the computing devices may be used. Various protocols such as Transmission Control Protocol/Internet Protocol (TCP/IP), Ethernet, File Transfer Protocol (FTP), Hypertext Transfer

Protocol (HTTP) and the like may be used, and the system can be operated in a client-server configuration to permit a user to retrieve web pages from a web-based server. Any of various conventional web browsers can be used to display and manipulate data on web pages.

(17) The disclosure is operational with numerous other computing system environments or configurations. Examples of computing systems, environments, and/or configurations that may be suitable for use with the disclosed embodiments include, but are not limited to, personal computers (PCs), server computers, hand-held or laptop devices, smart phones, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like that are configured to perform the functions described herein.

(18) The web browser **140** (e.g., a browser application) may be executed by the processor **103** of the computing device **101** and may receive code (e.g., hypertext markup language (HTML) code, JavaScript, etc.) from a server (e.g., a remote server that hosts a website, a local server that hosts an intranet site, and the like) and may execute the received code on the device for the benefit of a user of the computing device **101**. In some cases, the web browser **140** may include or be associated with one or more web applications (e.g., one or more of applications), where a web application may be configured to perform a single task or multiple tasks for a user. In such cases, the web application may be built using web technologies (e.g., JavaScript, cascading style sheet (CSS), HTML) and may be configured to be executed or interpreted by the web browser. Examples of web applications include games, photo editors, and video players that are run inside the browser.

(19) Web applications may be hosted applications or packaged applications. Hosted applications may include at least a portion of a web site that itself includes web pages, plus some metadata that may be especially pertinent to the web application or to the user of the web application to allow the web application to perform some particular functionality for the user. Packaged applications may be web applications whose code is bundled, so that the user can download all of the content of the web application for execution, for example by the browser. A packaged web application may not need to have network access to perform its functionality for the user, and rather may be executed (e.g., by the browser) locally on the computing device without access to a network. Packaged web applications have the option of using application programming interfaces (APIs) (e.g., extension APIs) that may allow packaged applications to change the way the browser behaves or looks. In some cases, the web browser **140** may include or be configured with various plug-in components, such as themes, backgrounds, or add-ons. A plug-in may be a set of software components that adds specific capabilities to a larger software application. A theme or background may include a favorite image and color scheme for a browser application.

(20) In various implementations, the web browser may include or be configured to interact with one or more browser extensions (e.g., the browser extension **150**). The browser extension **150** may include executable code packaged or grouped together as a definable whole, and configured to extend the functionality to the web browser. Thus, extensions also provide extra functionality to a browser, but generally, unlike web applications, there is little or no user interface component to the functionality provided by a browser extension. Instead, browser extensions extend the functionality of the browser and the web sites being viewed in it. For example, browser extensions can extend the functionality of the browser by adding a new button to the address bar, such as a language translator. Buttons like this can also apply to the current web site being viewed—for example, clicking the translation button can convert all text, or selected text, on the website presented to a user into a language chosen by the user. In another example, an extension can be installed so that when a user hovers a cursor over a thumbnail image on a webpage executed by a browser a larger-size version of the image is displayed until the user moves the cursor away from the image. In another example, an extension may be installed to embed a “share” button (e.g., a social media sharing button) on a page or near a sharable link. Compared to web applications, extensions cut

across websites and web applications. Extensions may be in effect across all websites (though some are site-specific). Web applications may not combine with other applications in this way. Rather, web applications generally run standalone, like any regular web site.

(21) When installed, browser extension code may often be viewed within a browser window and/or via another application, such as a text editor. As such, trade secret or other proprietary information enabling specific functionality of the browser extension **150** may be visible to a user. As such, a need has been recognized for a way to obfuscate the browser extension code to protect proprietary functionality, such as browser security functionality. For example, an enterprise organization may desire to integrate browser communication security functionality within a browser extension. In this illustrative example, a computing device, such as a personal computing device, may include a browser extension integrated with a web browser. The browser extension may be stored as an application and/or as source code on a computer-readable media device of the computing device. The source code of the browser extension may be executed by the web browser. While web browsers commonly permit the use of browser extensions to modify the functionality of the web browsers, existing browser extensions lack capabilities to detect “man in the middle” (e.g., “man in the browser”) interception from a malicious application (e.g., a middle agent) and prevent unauthorized data exposure to a third-party via the web browser. As such, a browser extension may be used to detect locally compromised client-server communications via a web browser and prevent exposure of sensitive information to an unknown third-party, as discussed in U.S. patent application Ser. No. 17/143,734 entitled “Web Browser Communication Validation Extension” and U.S. patent application Ser. No. 17/143,817, entitled “Browser Extension for Validating Communications”, both of which are incorporated by reference in their entirety.

(22) In the illustrative example of FIG. **1**, the browser extension **150** may be installed as extension code **151** in a format familiar to programmers and other users, such as HTML, CSS, JavaScript and the like. In some cases, a portion of the extension code may be used to provide a public functionality, such as the aforementioned “translate” button, while different (e.g., private or hidden) functionality, such as browser communication security functionality, may be obfuscated such that the code enabling private functionality is hidden from casual viewing.

(23) The extension code **151** may be installed, as a package, with one or more image files **157**. The image files **157** may include enterprise logos, background images, icons, or other such images that may be expected to be used with the browser extension **150**. For example, when installed and running, a browser extension may cause the web browser **140** to display the enterprise logo or other image as a background image for an open browser tab and/or a button image, a browser theme and/or the like. As such, the browser extension **150** may operate with a public functionality provided through the publicly viewable extension code **150** and displaying one or more of the image files **157**.

(24) As discussed herein, a steganographic file may contain content, such as image content audio content, and/or the like, where data is surreptitiously embedded within the file's image and/or audio content. For example, a common approach is to embed hidden data within an image of an image file (e.g., a portable network graphics (PNG) file, a joint photographic experts group (JPEG) file, and the like). In an illustrative method, the least significant bit of a pixel may be overwritten by an encrypted stream of secondary steganographic bits such that the primary public content of the image is not destroyed and/or distorted with notable artifacts. In some cases, the steganographic files may include additional information in metadata, comments, and the like. For example, PNG files and PEG files may contain comments where additional information corresponding to the image or image file is stored. Often, such comments may be used to store copyright, location or other information corresponding to the image or image file.

(25) Referring to FIG. **1**, one or more of the image files **157** may include code encrypted using, for example, steganography, such as by using least significant bit (LSB) encoding. For example, FIG. **5** shows an illustrative example of steganographic information included in an image. Other

steganographic encryption methods may be contemplated and/or other encryption methods may be used in addition to or instead of steganography. Also, while the image files **157** are illustrated, steganography may be used to encode information into other file types, such as audio files, text files, and the like. In some cases, comments or other metadata may be used to store additional executable code. These comments may be stored as simple text, encoded text corresponding to a binary or hexadecimal representation of characters. In some cases, the comments may also be encoded using LSB encoding to encode information in the least significant bit of a multiple characters in the comment of one or more of the image files **157**, such as and American standard code for information interchange (ASCII) character. In some cases, one or more image files may have encoded browser extension information or executable code encoded in a comment and/or steganographically encoded in the image information. When the comment information and/or the steganographic information is extracted and/or decoded, such as into the extracted code **153**, the browser extension **150** may execute the code to provide the private or hidden functionality of the browser extension.

(26) FIG. **2** shows an illustrative method **200** for processing obfuscated code for browser extensions in accordance with one or more aspects described herein. At **210**, browser extension code **151** may be loaded and processed to initiate operation of the browser extension **150**. The browser extension code **151** may include code to enable first functionality and may be stored in a form readable by a human. For example, the browser extension code **151** may include instructions that cause the browser extension **150** to provide functionality such as language translation for words or phrases presented on a web page, a visual theme for the browser extension, or other safe or innocuous functionality. In some cases, the browser extension code **151** may further load an image file at **220** to cause one or more images of the image files **157** to be displayed during normal operation. At **225**, when loaded, the browser extension code **151** may analyze each image file to determine whether one or more of the image files **157** includes a comment and/or corresponding information loaded in the comment. If a comment is identified, the comment may be extracted and/or decoded at **240** and a determination may be made if a last image has been identified at **235**. At **240**, the processor may process a function of the browser extension code **151** to perform the extraction and/or decoding of the comment information. In some cases, the browser extension code **151** used to extract the image comment may be visible to a user. In some cases, this browser extension code **151** may be obfuscated from casual viewing, for example, by redefining or overloading a function. When extracted and/or decoded, the comment information may be used as extracted code **153** capable of extracting steganographic or other encoded information from one or more of the image files **157**. In some cases, the steganographic or otherwise encoded information may be extracted from a same image file that included the comment. In some cases, the steganographic or otherwise encoded information may be extracted from a different image file that included the comment. In some cases, the steganographic or otherwise encoded information may be extracted from multiple image files of the image files **157**.

(27) If a comment is not associated with a particular image at **225**, then the processor **103** may determine whether a last image file has been examined at **235**. If not, the next image may be loaded at **220**. If the last image has been identified at **235**, the comment code may be processed to generate extracted code **153** at **250**. At **250**, the extracted code **153** from the comment may be processed to extract the steganographic information encoded in one or more of the image files **157**. In some cases, the extracted comment code may also extract a private key from a same or different image to facilitate extraction of steganographic information. At **260**, the extracted steganographic information may be processed into executable code and arranged into computer readable instructions capable of executing a non-public functionality of the browser extension **150**. For example, the non-public functionality of the browser extension **150** may correspond to browser security functions such as analyzing browser communications to identify a potential man-in-the-middle malware or other malware installed on the computing device **101**. At **270**, the browser

extension may be enabled to provide the public functionality (e.g., language translation, web browser theme functionality, and the like) and non-public functionality (e.g., security functionality) whose executable code is obfuscated from viewing by a human.

(28) FIG. 3 shows an illustrative operating environment **300** in which various aspects of the present disclosure may be implemented in accordance with one or more example embodiments. Referring to FIG. 3, a computing system environment **300** may be used according to one or more illustrative embodiments. The computing system environment **300** is only one example of a suitable computing environment and is not intended to suggest any limitation as to the scope of use or functionality contained in the disclosure. The computing system environment **300** should not be interpreted as having any dependency or requirement relating to any one or combination of components shown in the illustrative computing system environment **300**.

(29) The computing system environment **300** may include an illustrative browsing computing device **301** having a processor **303** for controlling overall operation of the browsing computing device **301** and its associated components, including a Random Access Memory (RAM) **305**, a Read-Only Memory (ROM) **307**, a communications module **309**, and a memory **315**. The browsing computing device **301** may include a variety of computer readable media. Computer readable media may be any available media that may be accessed by the browsing computing device **301**, may be non-transitory, and may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, object code, data structures, program modules, or other data. Examples of computer readable media may include Random Access Memory (RAM), Read Only Memory (ROM), Electronically Erasable Programmable Read-Only Memory (EEPROM), flash memory or other memory technology, Compact Disk Read-Only Memory (CD-ROM), Digital Versatile Disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed by the browsing computing device **301**.

(30) Although not required, various aspects described herein may be embodied as a method, a computing (e.g., data transfer) system, or as a computer-readable medium storing computer-executable instructions. For example, a computer-readable medium storing instructions to cause a processor to perform steps of a method in accordance with aspects of the disclosed embodiments is contemplated. For example, aspects of method steps disclosed herein may be executed by the processor **303** of the browsing computing device **301**. Such a processor may execute computer-executable instructions stored on a computer-readable medium.

(31) Software may be stored within the memory **315** and/or other digital storage to provide instructions to the processor **303** for enabling the browsing computing device **301** to perform various functions as discussed herein. For example, the memory **315** may store software used by the browsing computing device **301**, such as an operating system **317**, one or more application programs **319** (e.g., a web browser application), and/or an associated database **321**. In addition, some or all of the computer executable instructions for the browsing computing device **301** may be embodied in hardware or firmware. Although not shown, the RAM **305** may include one or more applications representing the application data stored in the RAM **305** while the browsing computing device **301** is on and corresponding software applications (e.g., software tasks) are running on the browsing computing device **301**.

(32) The communications module **309** may include a microphone, a keypad, a touch screen, and/or a stylus through which a user of the browsing computing device **301** may provide input, and may include one or more of a speaker for providing audio output and a video display device for providing textual, audiovisual and/or graphical output. The computing system environment **300** may also include optical scanners (not shown).

(33) The browsing computing device **301** may operate in a networked environment supporting connections to one or more remote computing devices, such as the computing devices **341** and **351**.

The computing devices **341** and **351** may be personal computing devices or servers that include any or all of the elements described above relative to the browsing computing device **301**.

(34) The network connections depicted in FIG. **3** may include a Local Area Network (LAN) **325** and/or a Wide Area Network (WAN) **329**, as well as other networks. When used in a LAN networking environment, the browsing computing device **301** may be connected to the LAN **325** through a network interface or adapter in the communications module **309**. When used in a WAN networking environment, the browsing computing device **301** may include a modem in the communications module **309** or other means for establishing communications over the WAN **329**, such as a network **331** (e.g., public network, private network, Internet, intranet, and the like). The network connections shown are illustrative and other means of establishing a communications link between the computing devices may be used. Various well-known protocols such as Transmission Control Protocol/Internet Protocol (TCP/IP), Ethernet, File Transfer Protocol (FTP), Hypertext Transfer Protocol (HTTP) and the like may be used, and the system can be operated in a client-server configuration to permit a user to retrieve web pages from a web-based server. Any of various conventional web browsers can be used to display and manipulate data on web pages.

(35) The disclosure is operational with numerous other computing system environments or configurations. Examples of computing systems, environments, and/or configurations that may be suitable for use with the disclosed embodiments include, but are not limited to, personal computers (PCs), server computers, hand-held or laptop devices, smart phones, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like that are configured to perform the functions described herein.

(36) FIG. **4** shows an illustrative block diagram of workstations and servers that may be used to implement the processes and functions of certain aspects of the present disclosure in accordance with one or more example embodiments. For example, an illustrative system **400** may be used for implementing illustrative embodiments according to the present disclosure. Referring to FIG. **4**, as illustrated, the system **400** may include one or more workstation computers **401**. The workstation **401** may be, for example, a desktop computer, a smartphone, a wireless device, a tablet computer, a laptop computer, and the like, configured to perform various processes described herein. The workstations **401** may be local or remote, and may be connected by one of the communications links **402** to a computer network **403** that is linked via the communications link **405** to the server **404**. In the system **400**, the server **404** may be a server, processor, computer, or data processing device, or combination of the same, configured to perform the functions and/or processes described herein. The server **404** may host local services and/or remote services. The server **404** may receive requests and send responses based on the protocol(s) associated with the local services and/or remote services.

(37) The computer network **403** may be any suitable computer network including the Internet, an intranet, a Wide-Area Network (WAN), a Local-Area Network (LAN), a wireless network, a Digital Subscriber Line (DSL) network, a frame relay network, an Asynchronous Transfer Mode network, a Virtual Private Network (VPN), or any combination of any of the same. The communications links **402** and **405** may be communications links suitable for communicating between the workstations **401** and the server **404**, such as network links, dial-up links, wireless links, hard-wired links, as well as network types developed in the future, and the like.

(38) One or more aspects of the disclosure may be embodied in computer-usable data or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices to perform the operations described herein. Generally, program modules include routines, programs, objects, components, data structures, and the like that perform particular tasks or implement particular abstract data types when executed by one or more processors in a computer or other data processing device. The computer-executable instructions

may be stored as computer-readable instructions on a computer-readable medium such as a hard disk, optical disk, removable storage media, solid-state memory, RAM, and the like. The functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents, such as integrated circuits, Application-Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects of the disclosure, and such data structures are contemplated to be within the scope of computer executable instructions and computer-usable data described herein.

(39) Various aspects described herein may be embodied as a method, an apparatus, or as one or more computer-readable media storing computer-executable instructions. Accordingly, those aspects may take the form of an entirely hardware embodiment, an entirely software embodiment, an entirely firmware embodiment, or an embodiment combining software, hardware, and firmware aspects in any combination. In addition, various signals representing data or events as described herein may be transferred between a source and a destination in the form of light or electromagnetic waves traveling through signal-conducting media such as metal wires, optical fibers, or wireless transmission media (e.g., air or space). In general, the one or more computer-readable media may be and/or include one or more non-transitory computer-readable media.

(40) As described herein, the various methods and acts may be operative across one or more computing servers and one or more networks. The functionality may be distributed in any manner, or may be located in a single computing device (e.g., a server, a client computer, and the like). For example, in alternative embodiments, one or more of the computing platforms discussed above may be combined into a single computing platform, and the various functions of each computing platform may be performed by the single computing platform. In such arrangements, any and/or all of the above-discussed communications between computing platforms may correspond to data being accessed, moved, modified, updated, and/or otherwise used by the single computing platform. Additionally or alternatively, one or more of the computing platforms discussed above may be implemented in one or more virtual machines that are provided by one or more physical computing devices. In such arrangements, the various functions of each computing platform may be performed by the one or more virtual machines, and any and/or all of the above-discussed communications between computing platforms may correspond to data being accessed, moved, modified, updated, and/or otherwise used by the one or more virtual machines.

(41) Aspects of the disclosure have been described in terms of illustrative embodiments thereof. Numerous other embodiments, modifications, and variations within the scope and spirit of the appended claims will occur to persons of ordinary skill in the art from a review of this disclosure. For example, one or more of the steps depicted in the illustrative figures may be performed in other than the recited order, one or more steps described with respect to one figure may be used in combination with one or more steps described with respect to another figure, and/or one or more depicted steps may be optional in accordance with aspects of the disclosure.

Claims

1. A computing device, comprising: at least one processor; a communication interface communicatively coupled to the at least one processor; and memory storing computer-readable instructions that, when executed by the at least one processor, cause the computing device to: load, by a web browser, first executable code of a browser extension; process, by the web browser, the first executable code to load a file, wherein the file comprises file content and metadata; decode, by the web browser, second executable code from the metadata of the file; extract, by the web browser by processing the second executable code, obfuscated information embedded in the file content; decode, by the web browser, the obfuscated information to extract third executable code; and

provide, by the web browser, first browser extension functionality by processing the first executable code and second browser extension functionality by executing the third executable code.

2. The computing device of claim 1, wherein the instructions, when executed by the at least one processor, cause the web browser to determine whether the file contents comprise encoded steganographic information.

3. The computing device of claim 1, wherein the instructions, when executed by the at least one processor, cause the computing device to extract, by the web browser processing the second executable code, the obfuscated information in the file content by causing the web browser to extract steganographic information from an image file using a least significant bit method.

4. The computing device of claim 1, wherein the file comprises an image file.

5. The computing device of claim 1, wherein the file comprises an audio file.

6. The computing device of claim 1, wherein the first executable code is visible to a user of the computing device.

7. The computing device of claim 1, wherein the third executable code is obfuscated to be unviewable by a user of the computing device.

8. A method, comprising: loading, by a web browser, first executable code of a browser extension; processing, by the web browser, the first executable code to load a file, wherein the file comprises file content and metadata; decoding, by the web browser, second executable code from the metadata of the file and from comments in the file content; extracting, by the web browser by processing the second executable code, obfuscated information embedded in the file content; decoding, by the web browser, the obfuscated information to extract third executable code; and providing, by the web browser, first browser extension functionality by processing the first executable code and second browser extension functionality by executing the third executable code.

9. The method of claim 8, comprising determining whether the file contents comprise encoded steganographic information.

10. The method of claim 8, comprising extracting, by the web browser processing the second executable code, the obfuscated information in the file content by causing the web browser to extract steganographic information from an image file using a least significant bit method.

11. The method of claim 8, wherein the file comprises an image file.

12. The method of claim 8, wherein the file comprises an audio file.

13. The method of claim 8, wherein the first executable code is visible to a user of a computing device.

14. The method of claim 8, wherein the third executable code is obfuscated to be unviewable by a user of a computing device.

15. Non-transitory computer-readable media storing instructions that, when executed by a computing device comprising at least one processor, memory, and a communication interface, cause the computing device to: load, by a web browser, first executable code of a browser extension; process, by the web browser, the first executable code to load a file, wherein the file comprises file content and metadata; decode, by the web browser, second executable code from the metadata; extract, by the web browser by processing the second executable code, obfuscated information embedded in the file content; decode, by the web browser, the obfuscated information to extract third executable code; and provide, by the web browser, first browser extension functionality by processing the first executable code and second browser extension functionality by executing the third executable code.

16. The non-transitory computer readable media of claim 15, wherein the instructions, when executed by the at least one processor, cause the web browser to determine whether the file contents comprise encoded steganographic information.

17. The non-transitory computer readable media of claim 15, wherein the instructions, when

executed by the at least one processor, cause the computing device to extract, by the web browser processing the second executable code, the obfuscated information in the file content by causing the web browser to extract steganographic information from an image file using a least significant bit method.

18. The non-transitory computer readable media of claim 15, wherein the file comprises an image file.

19. The non-transitory computer readable media of claim 15, wherein the file comprises an audio file.

20. The non-transitory computer readable media of claim 15, wherein the third executable code is obfuscated to be unviewable by a user of the computing device.
